



Improving Dairy Farmers' Income through Scientific Calf Rearing





Farmers' perspectives on induction of new dairy animals from outside

Is animal
producing as
per
expectation?

Milk let-down
problem?



Health
problem?



Do we have dairy farm record book?



**Why keep
records?**



Is traditional dairying profitable?

- Lactation length: 270 days
- Dry period: 150 days
- Lactation yield: 1568 liters
- Sale price of milk per liter: Rs. 29.52 (4.8% fat)

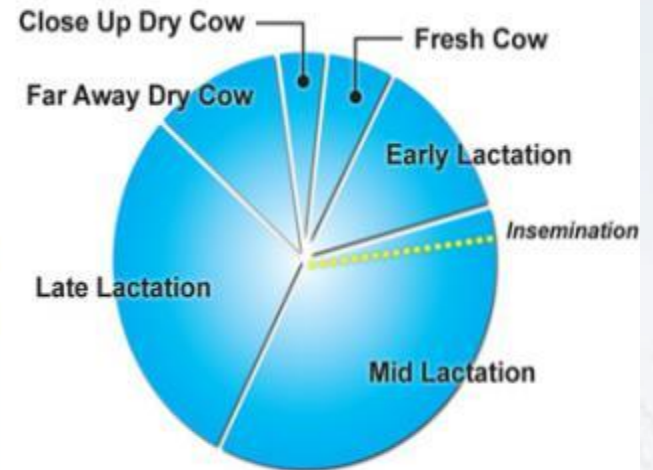
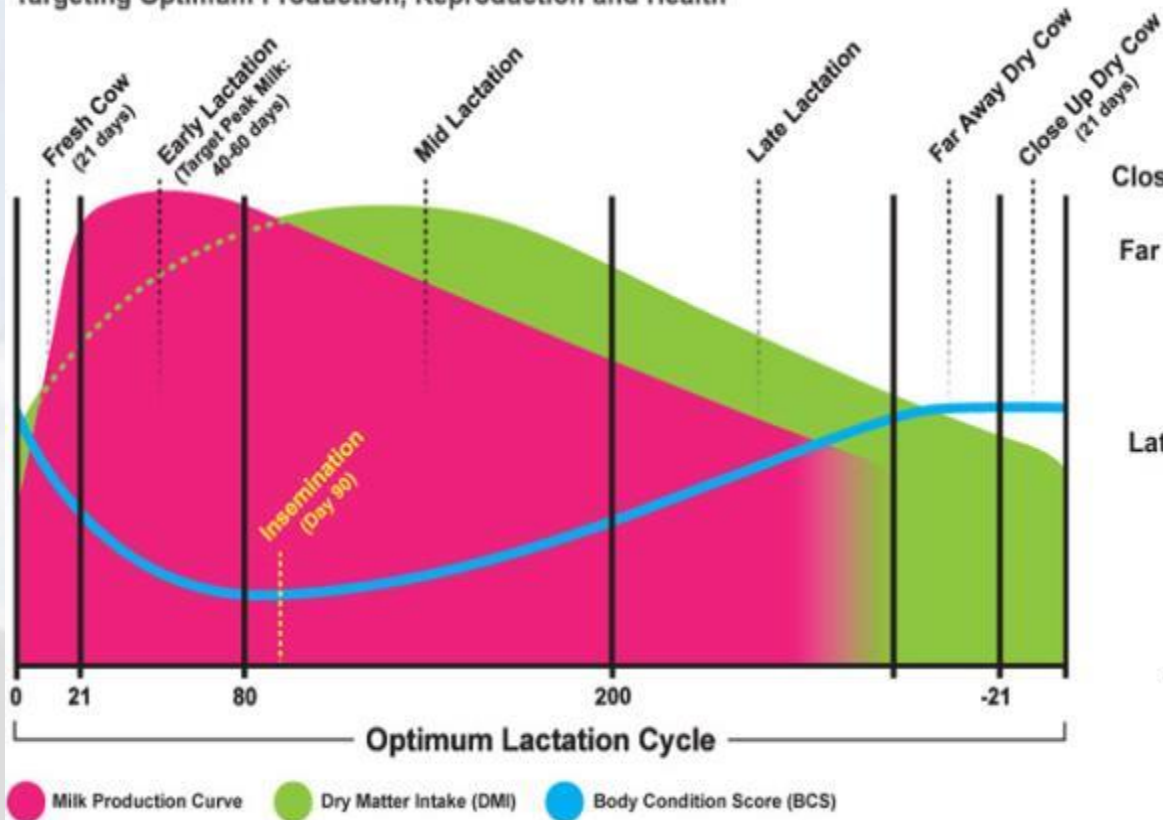
Particulars	Rate per kg (Rs)	During lactation period		During the dry period	
		Kankrej cow		Kankrej cow	
		Qty. in (kg)	Cost in (Rs)	Qty. in (kg)	Cost in (Rs)
Green fodder	1.5	7	10.5	7	10.5
Dry fodder	5	7	35	7	35
Concentrates	17.25	4	69	1	17.25
Total cost (Rs.)			114.5		62.75

Sl. No.	Particulars	Kankrej Cow (Rs.)
I	Fixed cost	5000
II	Recurring expenditure	
	Feeding during lactation period	30915
	Feeding during the dry period	9412
	Veterinary aids	300
	Labour charges	10000
	Total (Fixed + Recurring)	55627
III	Income	
	(a) Sale of milk	46287
	(b) Sale of manure	2570
IV	Profit/Loss	-6770

The Lactation Cycle

The Lactation Cycle

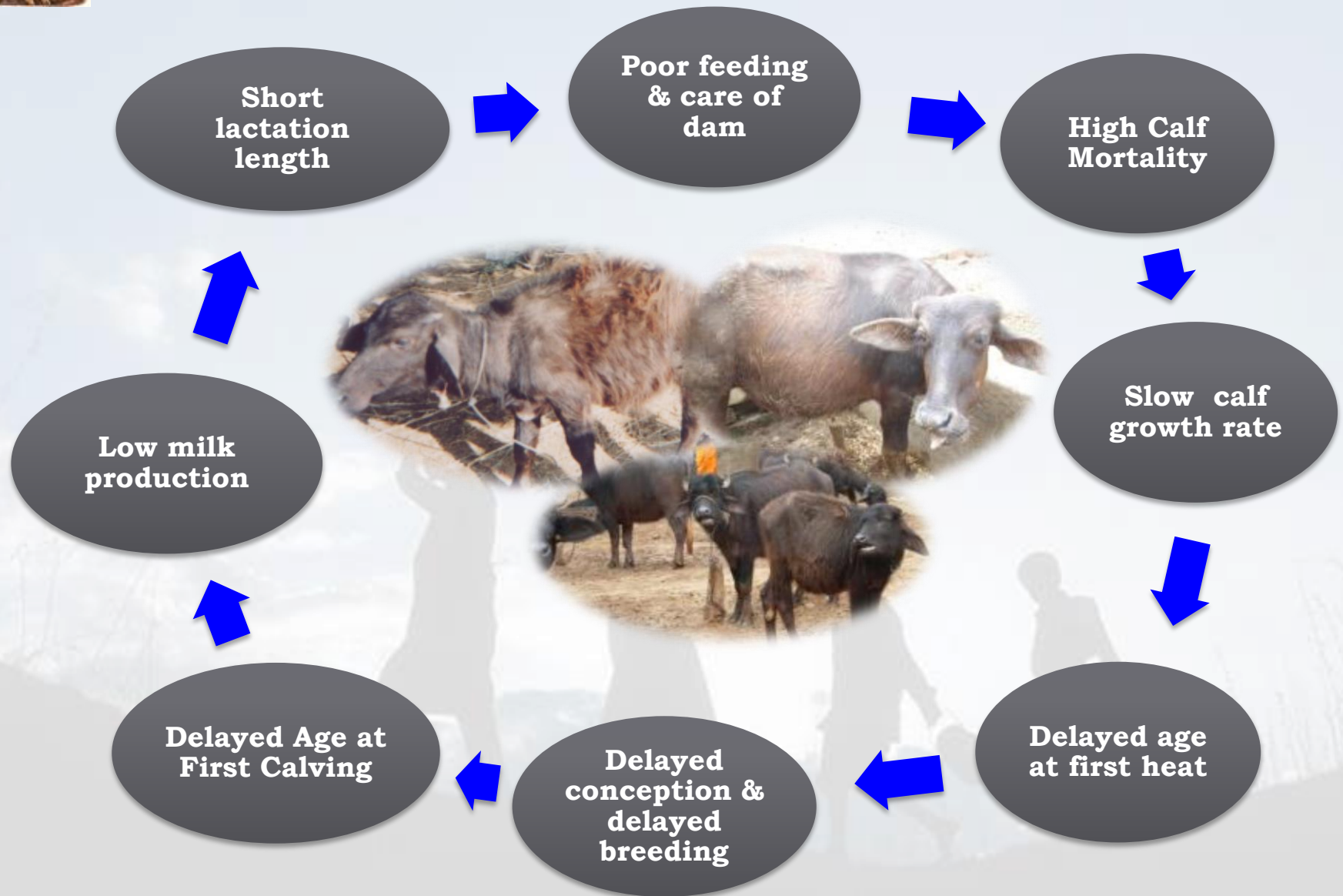
Targeting Optimum Production, Reproduction and Health



"What happens today is a result of what happened yesterday"

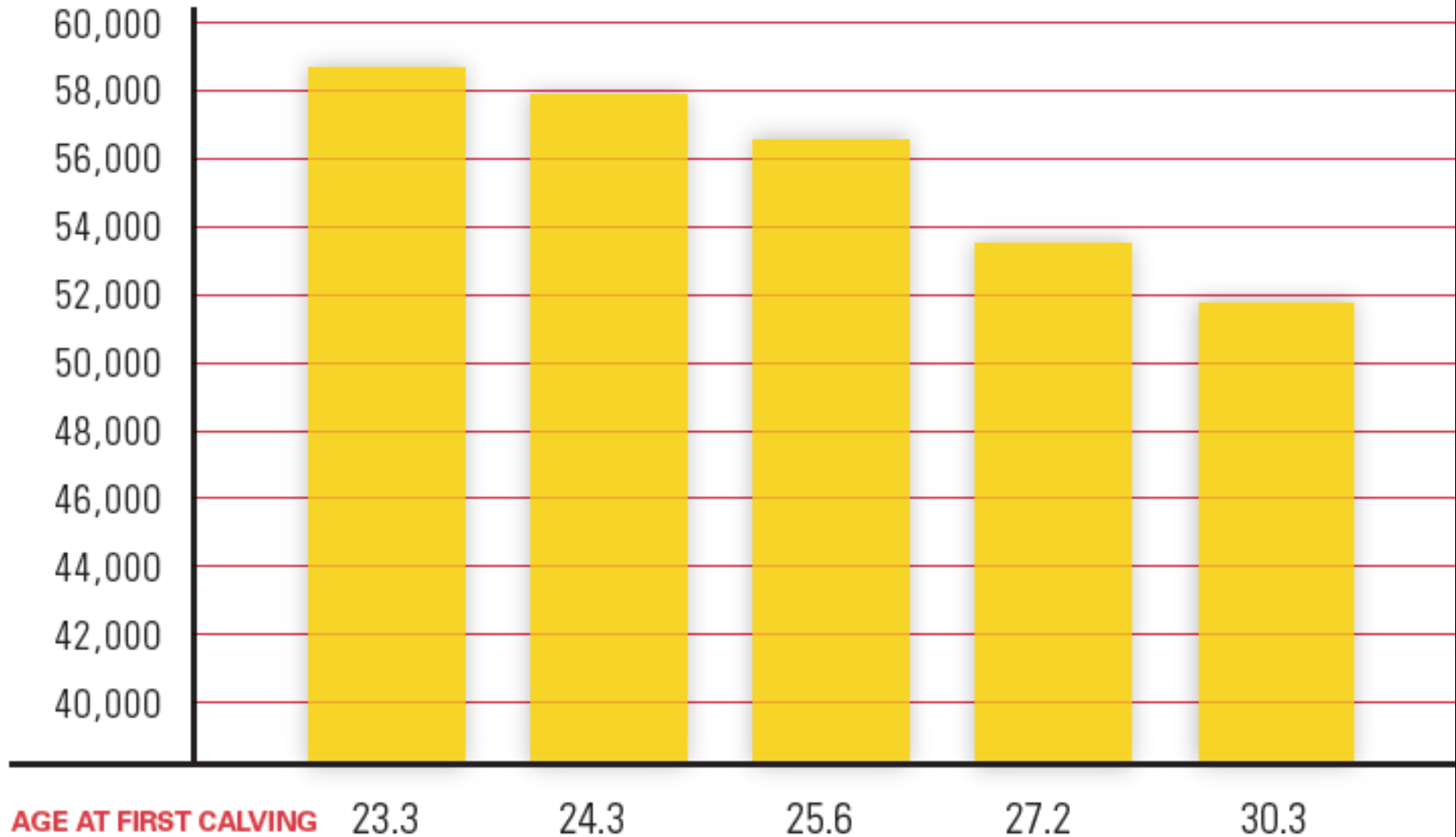


Bottlenecks of current rearing practices



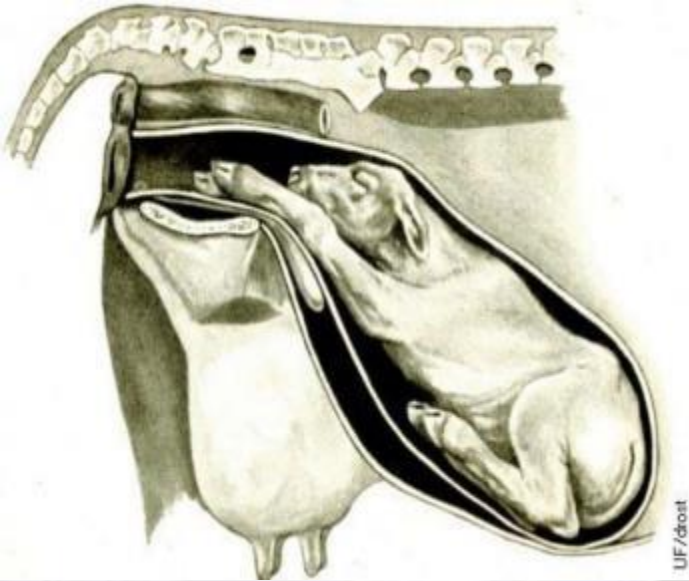
Age at first calving and lifetime milk production

LIFETIME MILK PRODUCTION





NDDB's Calf Rearing Programme



Prenatal



Neonatal



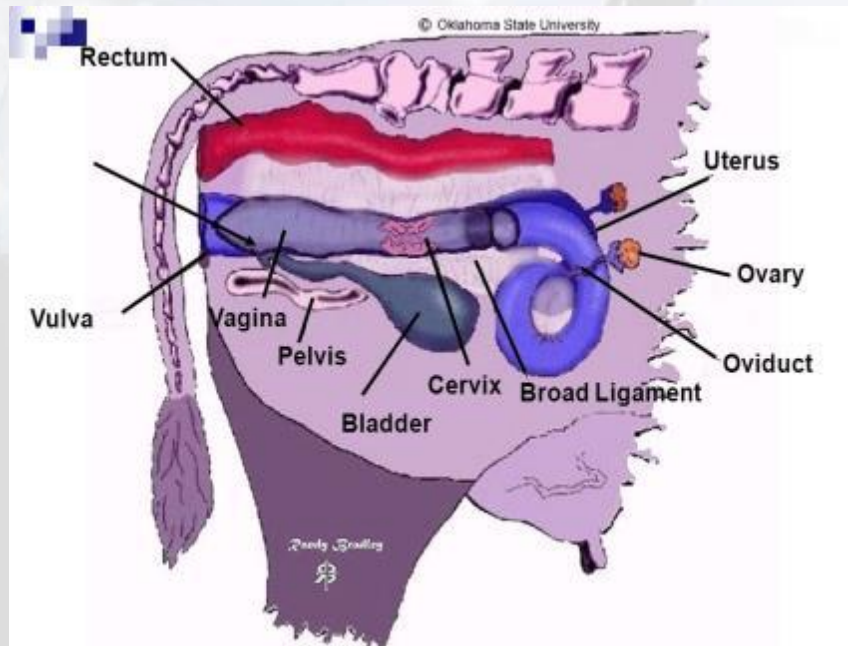
Postnatal



Pregnancy feed – Salient features



- **Designed to address Dietary Cation-Anion Balance.**
- **‘Highly nutritious feed’ - high protein, high energy, coated vitamins, chelated minerals, right protein-energy ratio to account for**
‘reduced feed intake in pregnant animals due to reduced rumen size.’





Calf management at birth

The first hour after calving is the golden hour for calf



Clean nostril and mouth and allow the mother to lick



Cut the naval cord and dipping into 7% tincture iodine



Feeding 2 ltrs. Colostrum within 1-2 hrs of birth - **“a passport to life”**



De-worming within 10-14 days of age

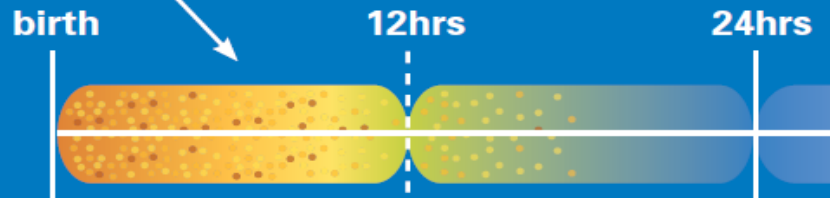
Antibodies are only absorbed for a short time...



Minimum recommendation

At least 2 litres of good quality colostrum in the first 12 hours.

Then, at least another 2 litres of good quality colostrum within the next 12 hours.



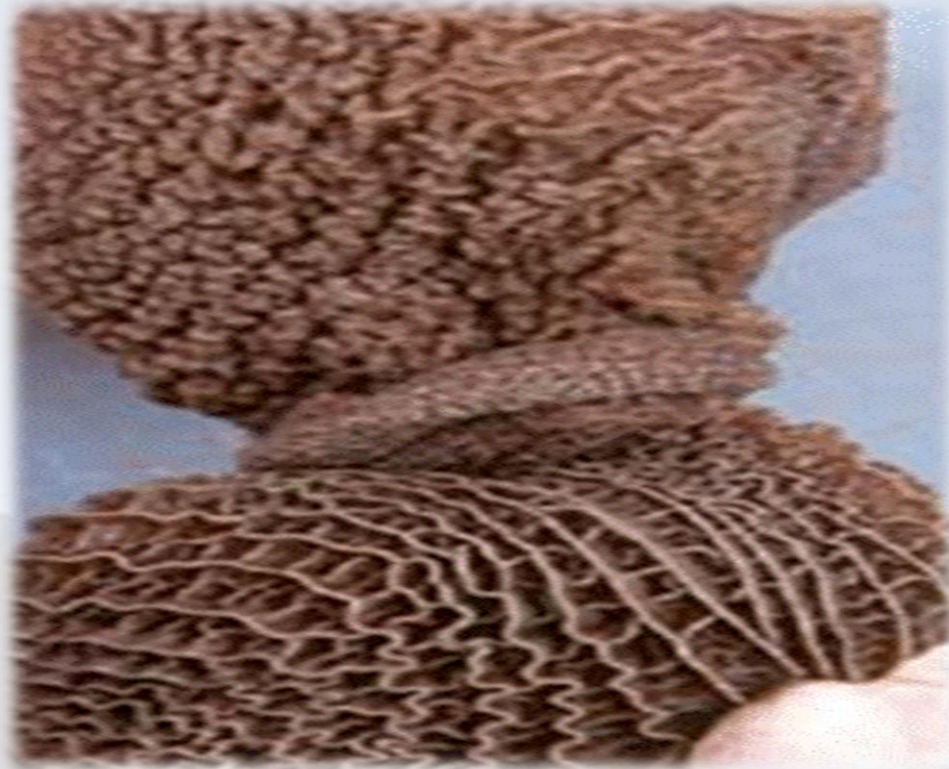
You may get even better results by giving larger quantities of quality colostrum earlier.
Use between 10–12% of body weight as a guide.



Calf Starter – Salient Features



**Underdeveloped rumen
(Calf on traditional feeding)**



**Well developed rumen & papillae
(Calf fed with calf starter)**

Formulated taking into account Butyrates, Antioxidants, Mold inhibitors, Available Phosphorus, Strict Aflatoxin control and Zero NPN content.



The rumen epithelium of 6 week old calves fed different rations

Scientific Calf Rearing Programme

(A modular approach)

Implementation Steps

- Selection of villages
- DCS committee meeting
- Selection of field supervisors & training
- VAP
- Implementation in field

Objective of Initiative :

- To reduce age at maturity and age at first calving
- Reduce calf mortality by 50%
- Increasing lifetime productivity of animals.

Monthly Follow up





Calf feeding schedule (birth to 26th wk age)

	Colostrum/ Whole Milk (kg/day)	Calf starter (kg/day)	Good quality hay* (kg/day)	Green fodder* (kg/day)
0-2 days	1.5 to 2.0 colostrum			
3-4 days	1.5-2.0 milk			
4-14 days	1.0-1.5 milk	0.10	0.10	
3 rd week	0.5-1.0 milk	0.20	0.15	0.15
4 th week	<i>Milk (0.5 kg) or milk replacer (0.25 kg) can be fed, if available with progressive dairy farmers & having good economic conditions</i>	0.25	0.20	0.25
5 th week		0.40	0.30	0.40
6 th week		0.50	0.40	0.50
7 th week		0.60	0.60	0.60
8 th week		0.70	0.80	0.70
9 th week		0.80	0.90	0.80
10 th -11 th wk		1.00	0.90	1.00
12 th week		1.20	1.00	1.00
13 th -16 th wk		1.50	1.20	1.20
17 th -20 th wk		1.75	1.50	1.50
21 st -26 th wk		2.00	1.50	1.50-2.0

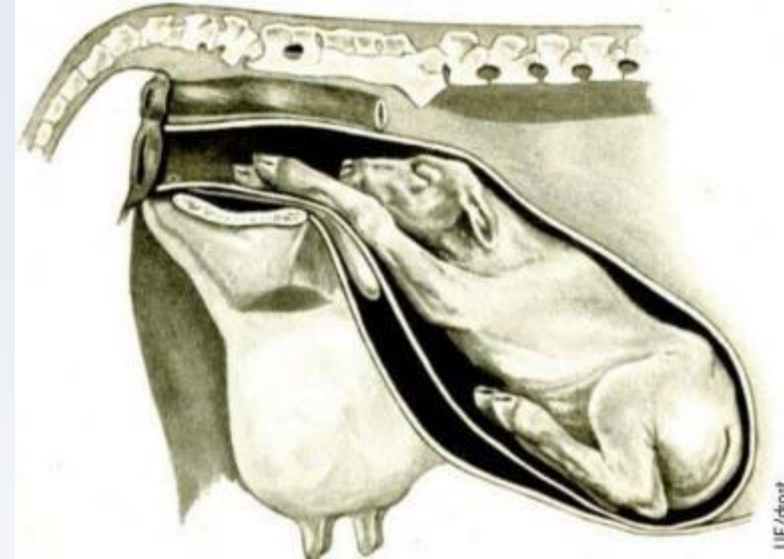
Note: *Requirement of hay and green fodder may vary from breed to breed & body weight of calf. Colostrum feeding is very essential during early life of calf.

Note: 27th to 108 week of age, calf growth meal @ 2.0kg /day will be fed to calves.



Golden rules for rearing healthy calves

- Start the feeding desired nutrient right from the advanced pregnant stage to give birth to healthy calf.
- Provide adequate quantity of colostrum within 2 to 3 hours of birth.
- Take preventative action for navel and respiratory infections.
- Ensure proper clean and dry housing and bedding
- Take care of the calves nutritional requirements like feeding of milk and calf starter
- Proper de-worming and vaccination schedule must be followed





Effect of feeding calf starter/growth meal on daily weight gain



Control

Age: 6 months

Body wt: 91 kg

Average daily gain: 325 g/day



Experimental

Age: 6 months

Body wt: 123 kg

Average daily gain: 498 g/day



Effect on sexual maturity

Control



First heat: at 25 months

Body wt: 236 kg

Experimental



First heat: at 17 months

Body wt: 250 kg



Average daily gain of Kankrej calves

Control



Treatment



Body weight at 3 months: 54 kg

ADG (g/d): 300 g/d

Body weight at 3 months: 75 kg

ADG (g/d): 500 g/d



Effect of feeding calf starter/growth meal on daily weight gain

Control

Treatment



Body weight at 5 months: 70 kg

ADG (g/d): 313 g/d

Body weight at 5 months : 110 kg

ADG (g/d): 566 g/d



Tag ID: 340143261753

Date of Birth: 03/03/2017 (13 months)

Birth Weight: 30kg

Present Weight: 271kg

ADG (g/d): 617g/day

Village: Nokha (Banaskantha)

Farmer's Name: Danobhai Ranabhai Patel

Exhibited Heat at: 11 months





Buffalo calves - First heat at 11.5 months age

Tag Number:
130007416590

Birth weight - 40kg
DOB - 6 June, 2017
Present Body wt. - 310kg
ADG= 780g/day

Owner: Mastan Singh
Village: Kheri Gurna
(Ropar Milk Union)





Tag ID: 340095523570

Date of Birth: 12/2/2017 (14 months)

Birth Weight: 25kg

Present Weight: 291kg

ADG (g/d): 633g/day

Village: Madal

Farmer's Name: Patel Valabhai Vagtaji

Exhibited Heat at: 13 months





Tag ID: 340144585782

Date of Birth: 3/3/2017 (13 months)

Birth Weight: 26kg

Present Weight: 250kg

ADG (g/d): 574g/day

Village: Madal

**Farmer's Name: Patel
Harchandbhai Punamji**

Exhibited Heat at: 12 months





Tag ID: 340144603901

Date of Birth: 2/3/2017 (13 months)

Birth Weight: 32kg

Present Weight: 226kg

ADG (g/d): 497g/day

Village: Madal

**Farmer's Name: Patel
Hardanbhai Vardhaji**

Exhibited Heat at: 13 months





Buffalo calves under CRP Punjab



Control

Vill: Maavi Kalan
Age: 2.5 months
Body weight: 59kg
Tag ID:130005494196
Date of calving: 27.09 .2017
Nirbhay Singh



Experiment

Vill: Salana
Age: 2 months
Body weight: 80kg
Tag ID:130007285860
Date of calving: 17.10 .2017
Amrinder Singh



Growing buffalo calves under CRP Punjab



Control

Vill: Maavi Kalan
Age: 5 months
Body weight: 82kg
Tag ID:130007267060
Date of calving: 24.08 .2017
Jagseer Singh



Experiment

Vill: Kheri Gurna
Age: 5.5 months
Body weight: 234kg
Tag ID:130001281707
Date of calving: 06.08.2017
Mastan Singh



Today's calf is tomorrow's cow...

**Thanks for
your love and
care..**